Higher National Diploma in Civil Engineering

First Year, First Semester Examination – 2019

CE1103- Engineering Surveying I

Instructions for Candidates:
Answer any **Five (05)** questions.
All question carry equal marks

No. of questions : 06
No. of pages : 03
Time : Three (03) hours

1.

i. Define "surveying" (02 marks)

ii. What are the fundamental principles of surveying? (04 marks)

iii. Differentiate between plane surveying and geodetic surveying (04 marks)

iv. Define the terms " vertical direction" and " horizontal direction" (04 marks)

v. Briefly describe any three types of specializes surveys. (06 marks)

(Total 20 marks)

2.

i. What is the meaning of "well-conditioned triangle"? (02 marks)

ii. Define the terms "offset" and "Dead measurement" (04 marks)

iii. What are the instruments used in chain surveying (03 marks)

iv. Briefly describe the operations (steps) involved in chain surveying (06 marks)

v. What are the advantages and disadvantages of chain surveying? (05 marks)

(Total 20 marks)

3.

i. Briefly describe the way of measuring 100m on an uneven ground (with the sketches) with mentioning the used instruments (05 marks)

ii. Briefly explain two way of dropping right angle (perpendicular offset) from a point to the chain line. (04 marks)

iii. While measuring a chain line AB, the following offsets are taken. Enter these details on a double line field book page.

a. A telegraph post is 2.5 m perpendicular from chainage 10 m to the right of the chain line.

b. A road crosses obliquely from left to right at chainage 10 m and 14 m. Perpendicular offsets are 2m and 3m to the side of the road from chainage 5m and 20 m respectively

c. A tube-well is 5m perpendicularly from chainage 30 m to the left of the chain line.

d. Total chainage of AB is 45 m. (08 marks)

iv. Mention three main considerations have to be taken on plotting. (03 marks)

(Total 20 marks)

4.

i. Briefly discuss the types of errors in chaining while obtaining the distance measurements. (06 Marks)

ii. Define the terms "Accuracy" and "precision". (04 marks)

iii. What do you understand by standard deviation & most probable value (04 marks)

iv. An angle "A" was measured by different persons and the following are the values.

Angle	Number of measurements
65° 30′ 10"	2
65° 29′ 50"	3
65° 30′ 00"	3
65° 30′ 20"	4
65° 30′ 10"	3

Find the most probable value of the angle. (06 marks)

(Total 20 marks)

5.

i. The downhill end of the 30 m (slope length) tape is held 80 cm too low (difference in height). What is the slope error and horizontal length? (05 marks)

ii. A distance of 1500 m was measured with a 20 m chain. After the measurement chain was found to be 80 mm longer. If the length of chain was perfectly correct while starting measurement, what is the true length of the line measured? (05 marks)

iii. A line was measured with a steel tape which was exactly 30 m at 18°C and a pull of 50 N and the measured length was 459.242 m. Temperature during measurement was 28°C

and the pull applied was 100 N. The tape was uniformly supported during the measurement.

Find the true length of the line if the cross-sectional area of the tape was 0.02 cm^2, The coefficient of thermal expansion per °C = 0.0000117 and the Young's modulus of elasticity is 21 $x10^6$ N per cm^2. (10 marks)

(Total 20 marks)

6.

i. Briefly explain the terms "field book" and" field notes". (04 marks)

ii. What are the information that should be included at the recording in handwritten field book? (03 marks)

iii. Briefly describe the types of EDM instruments which are classified based on the type of carrier wave (09 marks)

iv. Briefly explain the advantages and disadvantages available with EDM instrument while usage in distance measurements. (04 marks)

(Total 20 marks)